

AI Applications in Energy Analytics

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Introduction

What this talk covers

AI applications and tools around analytical models

1

AI-based forecasting: PSRCast

Integrated scenarios for renewables, inflows and demand — from days ahead to climate change

2

AI as tools

Grounded assistants (RAG), agents acting on SDDP data through MCP, and intelligent multi-agent orchestration

3

Blending optimization and AI

Reasoning agents that explore hypotheses to build operating policies, leading to autonomous energy workflows

Explore our Analytics Report!



Analytics Report

PSR

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Artificial intelligence is rapidly becoming a core capability in energy analytics, not only as a tool for automation, but as a new layer for modeling, reasoning, software development, and large-scale computation. This issue of the Analytics Report highlights how PSR is positioning itself at the forefront of AI applied to energy systems, combining climate scenario generation, stochastic optimization, reinforcement learning, agentic workflows, MCP-based interaction with analytical tools, AI-assisted coding, and GPU-based optimization. Together, the articles show how AI and advanced computing can support more robust methodologies, faster studies, and new ways of developing, operating, and interpreting complex energy models.



AI-Based Forecasting: PSRCAST

Forecasting under a changing climate

Why historical statistics are no longer enough

The problem

- The energy transition makes power systems weather-dominated: hydro, wind and solar
- "Climate change" changes the distributions of inflows, wind and temperature — and of extremes
- Models that reproduce historical patterns miss the new risk landscape
- GCM projections are abundant; the challenge is turning them into usable scenarios

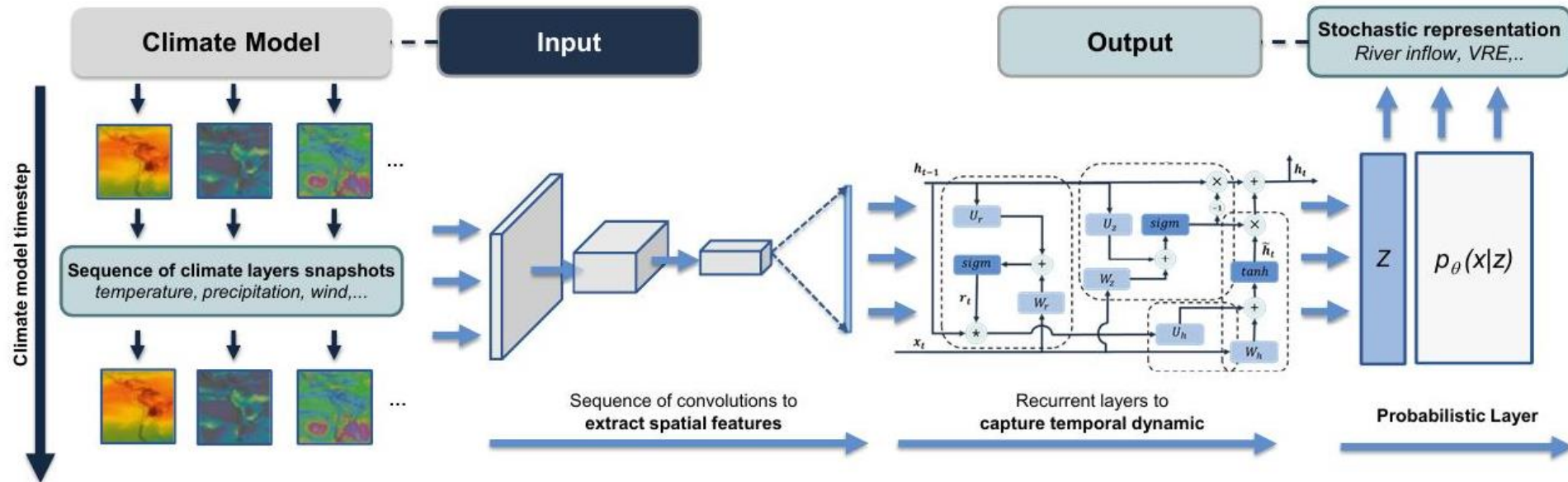
Our answer

- + An AI model that transforms projections from global circulation / weather models into plausible scenarios of inflows, renewables and demand
- + Probabilistic by construction: a robust scenario generator conditioned on global climate signals
- + Consistent across variables, locations and horizons
- + Feeds our stochastic planning models (SDDP and beyond)

From global climate signals to decision-ready stochastic scenarios.

How PSRCast works

Deep learning that emulates climate models, thousands of times faster



One engine, every horizon — fine-tuned per climate model, with no a priori bias correction or downscaling:

Up to 15 days

ECMWF-IFS, NOAA-GFS + AI weather models
updated daily

Up to 12 months

Seasonal: ECMWF-SEAS5, NOAA-CFSv2, NASA, GFDL, ...
updated monthly

Up to 10 years

Decadal: EC-Earth3, CMCC-CM2, HadGEM3, ...
updated yearly

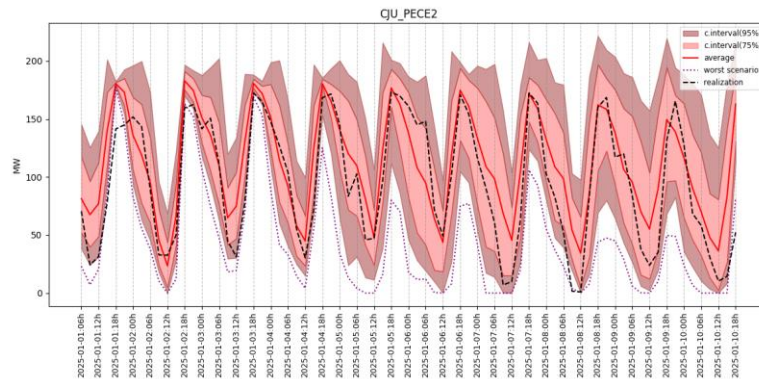
End of century

CMIP / ScenarioMIP ensemble (~40 GCMs)
per CMIP phase

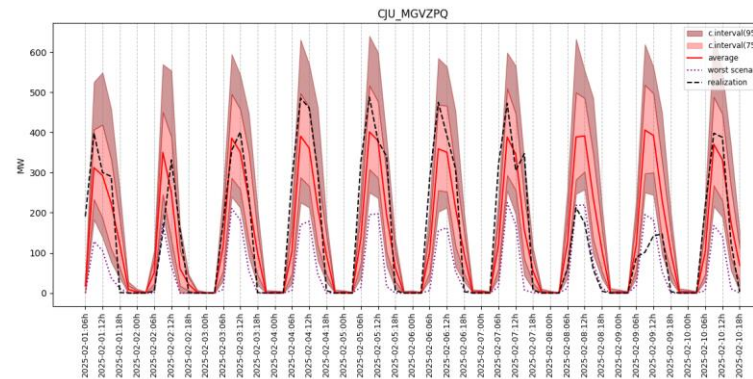
Short-term skill: out-of-sample validation

Probabilistic forecasts vs. what actually happened, 1–10 days ahead (Brazil)

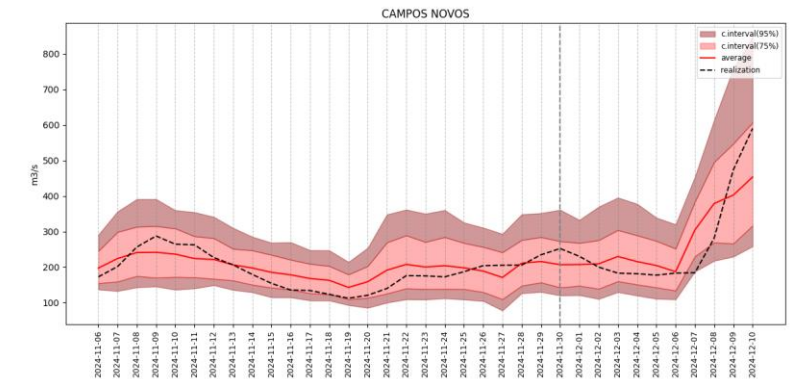
Wind — Pernambuco



Solar — Minas Gerais



River inflow — Campos Novos (Uruguai)



- + Shaded bands: 75% / 95% confidence intervals; dashed line: realized generation / inflow
- + Inflows are a dynamic system: the network carries a hidden hydrological state, initialized from recent observation

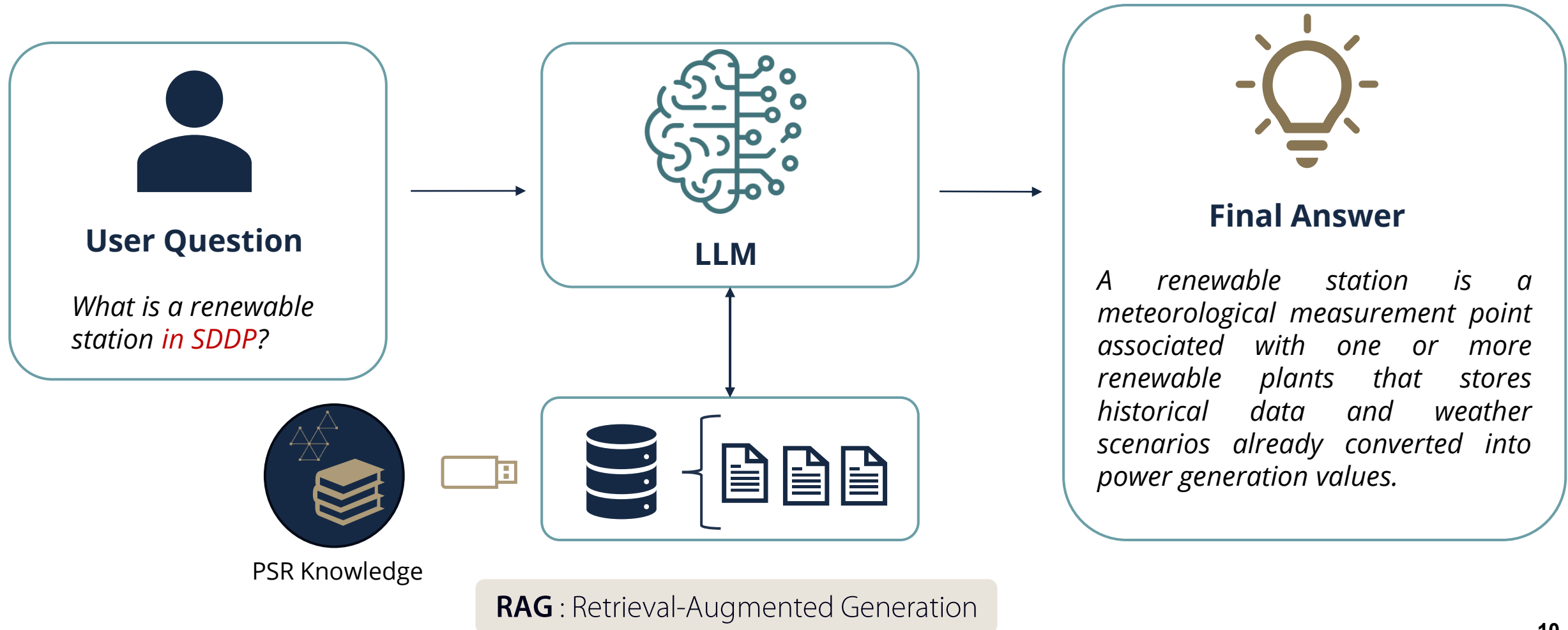
The model is a calibrated distribution of futures — not a single trajectory.



Grounded Intelligence

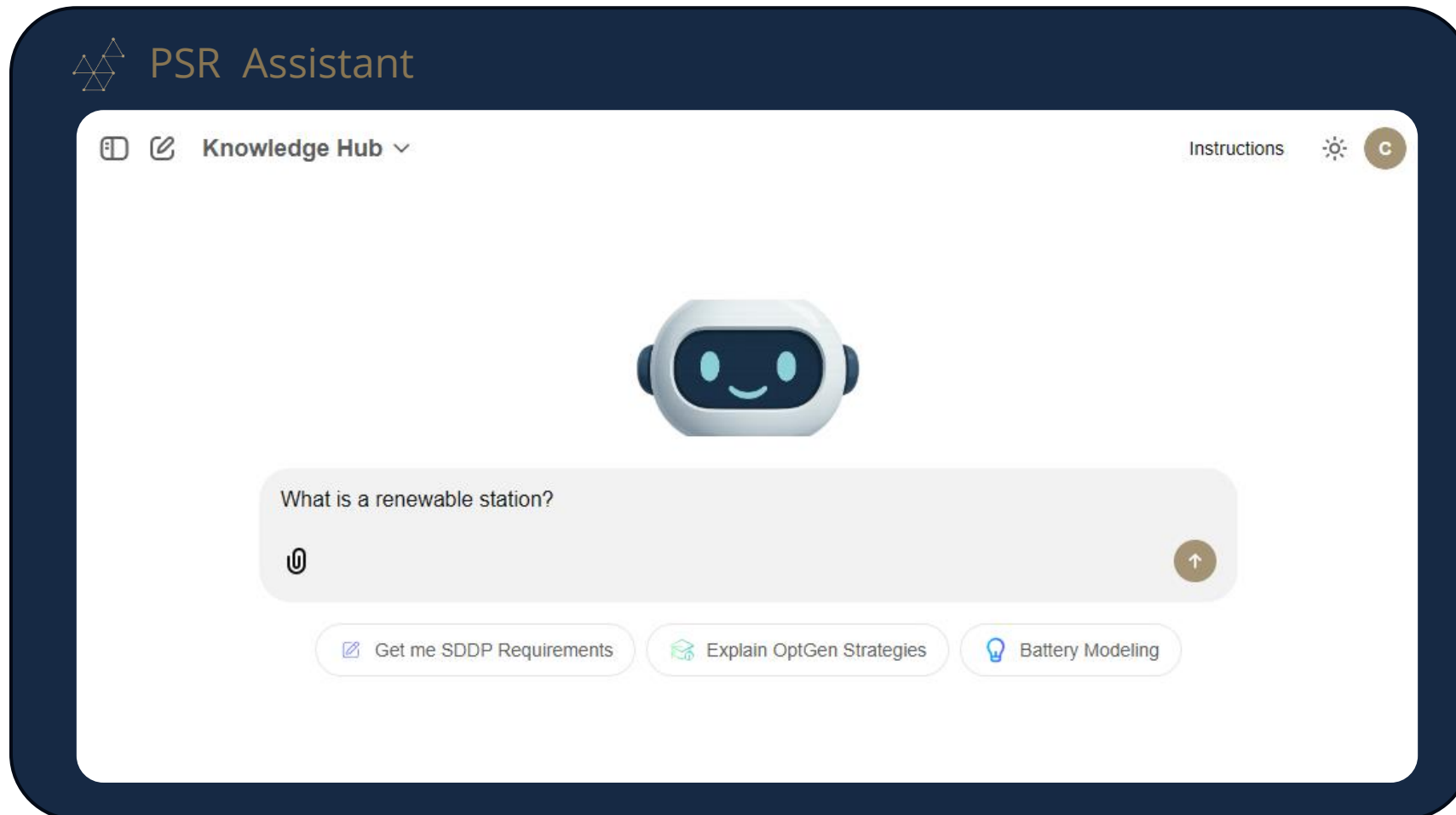
Building a Bridge Between LLMs and our Knowledge

Ensuring accuracy through RAG



Meet our Assistant

Accurate answers, grounded in our documentation



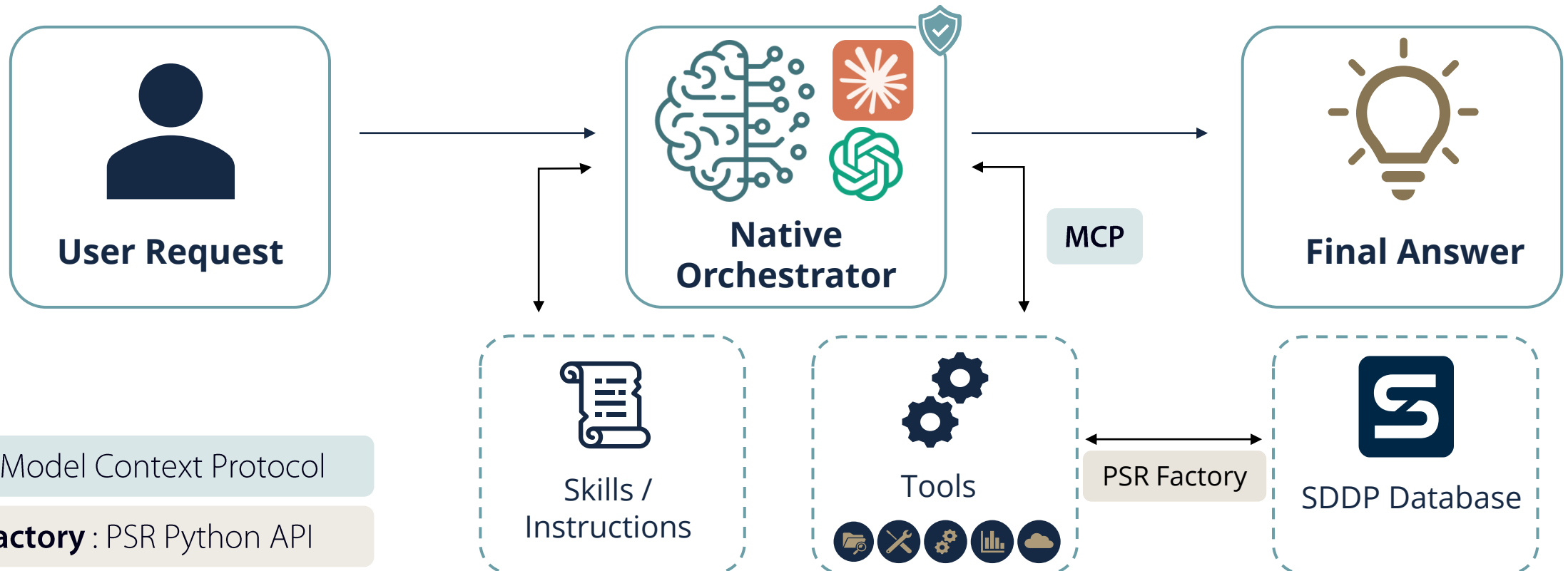


Agentic AI: From Answers to Actions

How Our Agents Access Your Data

Standardizing the connection between models and SDDP databases

An agentic architecture is a system designed to **execute complex workflows** by combining autonomous planning, long-term memory, and the ability to **interact with external applications**.



PSR Cloud Integration

1. Get case status
2. Download Results and plot Annual Marginal Cost

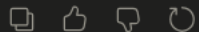
Can you please check the status of the run with ID 678559 on PSR Cloud?

Loaded tools, used psr-cloud integration >

The run with ID 678559 has completed successfully! Here's the summary:

- **Case ID:** 678559
- **Status:** ☒ SUCCESS
- **Message:** Finished successfully

The results should be available for download if you need them. Would you like me to retrieve the case files or check the execution log?



Please download the SDDP.html file for this case from PSR Cloud and plot the annual marginal cost graph.

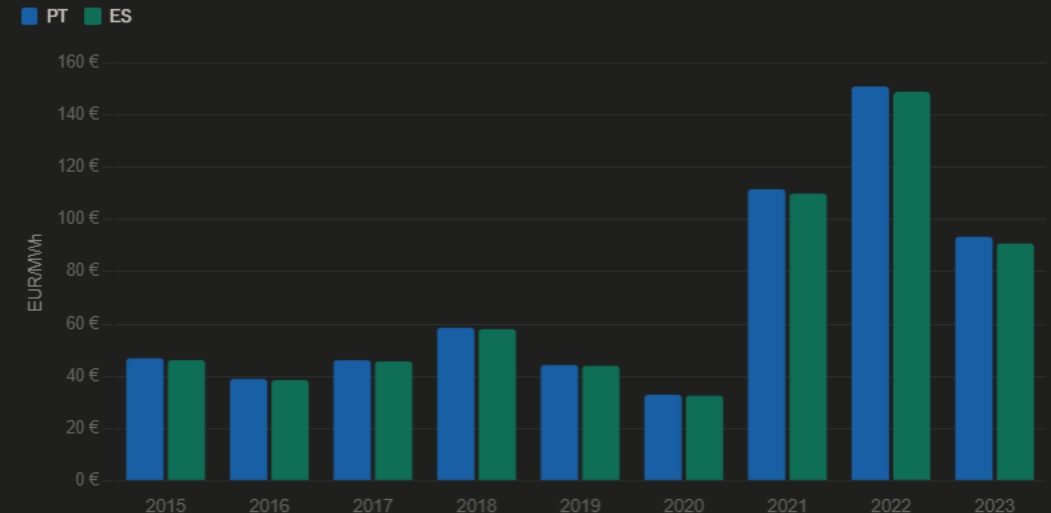
Used 3 tools >

Peak year
2022

PT peak (2022)
151.1 EUR/MWh

ES peak (2022)
149.1 EUR/MWh

Lowest year
2020



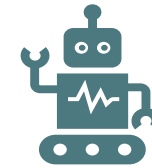
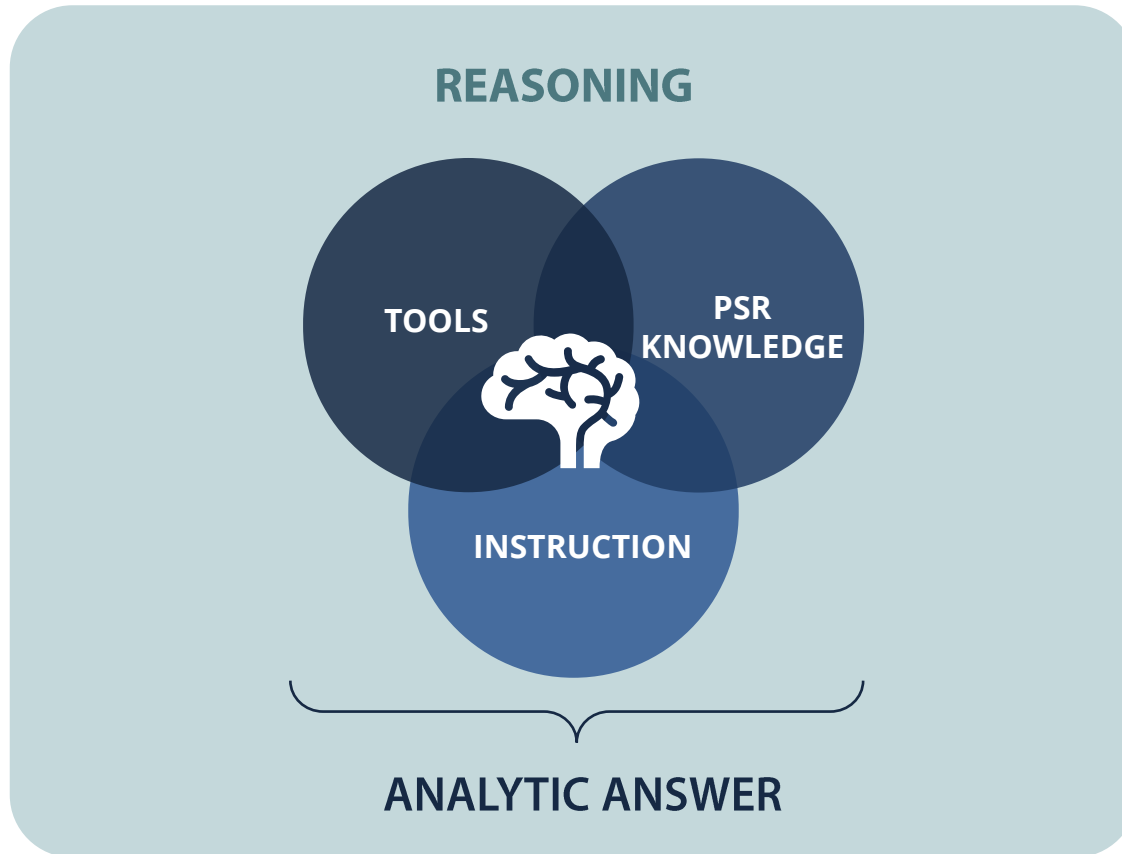
Here's the annual marginal cost chart from case 678559, showing both the PT (Portugal) and ES (Spain) systems from 2015 to 2023 in EUR/MWh.





Intelligent Orchestration: The PSR AI Brain

The Thinking Machine: Autonomous Analysis



Multi-Agent Orchestration

Orchestrates **specialized agents** to analyze **SDDP/NCP cases**



Intelligent Result Interpretation

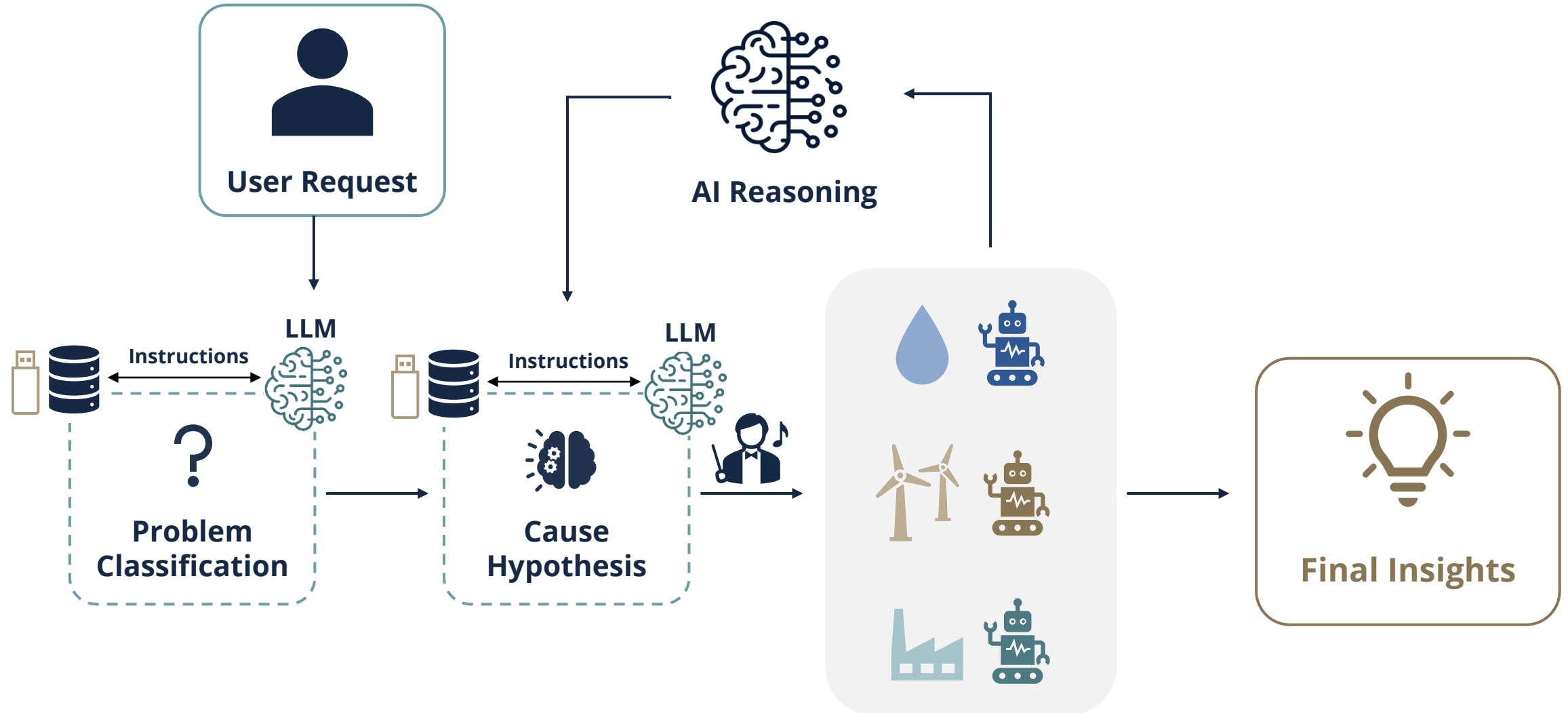
Turning **SDDP outputs** into actionable **insights** for **decision-making**



Evidence-Based Synthesis

Produces **evidence-backed conclusions** to support **strategic decisions**

From Data to Insight: Multi-Agent Pipeline





Blending Optimization and AI

High-Level Prompt Given to the Agent

You are a hydrothermal system operation specialist...

Your task is to build an operating policy for the stochastic system described in...

Through a custom MCP service..., you have access to domain tools that allow you to...

Reason like an experienced planning engineer analyzing the system before making decisions.

Expected reasoning process:

Understand the system: Identify reservoirs, storage capacities, initial conditions...

Baseline diagnosis: Simulate the system without additional cuts and analyze myopic behavior, reservoir depletion...

Sensitivity analysis: Experiment with specific conditions using local simulations to estimate the future value of stored water....

Cut construction: Propose stage-level future-cost cuts representing the opportunity value of water...

Validation and refinement: Re-simulate the system, evaluate policy behavior, and iteratively refine the cuts.



Sonnet 4.6 ▾

 inDepth

ASSESSING THE CAPABILITY OF AGENTIC AI IN A STOCHASTIC HYDROTHERMAL SCHEDULING CASE STUDY

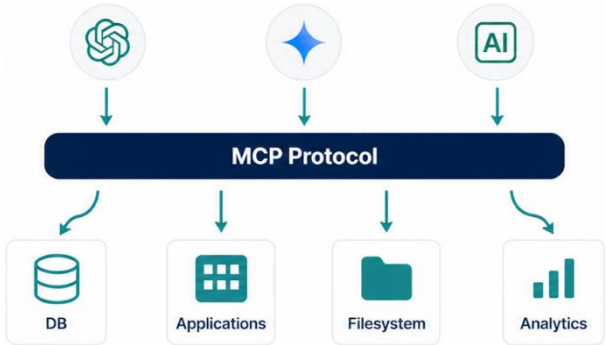
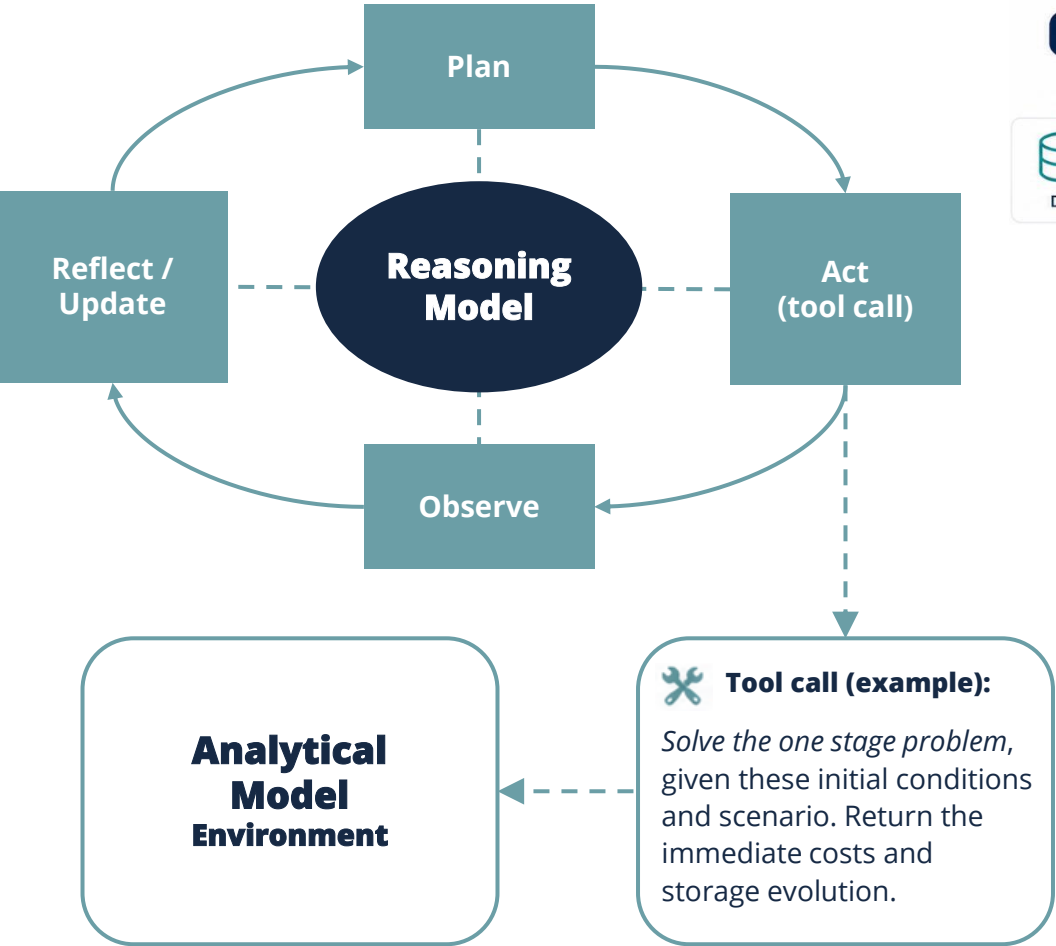
Julio Alberto Silva Dias and Mario Veiga Ferraz Pereira

Abstract



Framework and Results of the Experiment

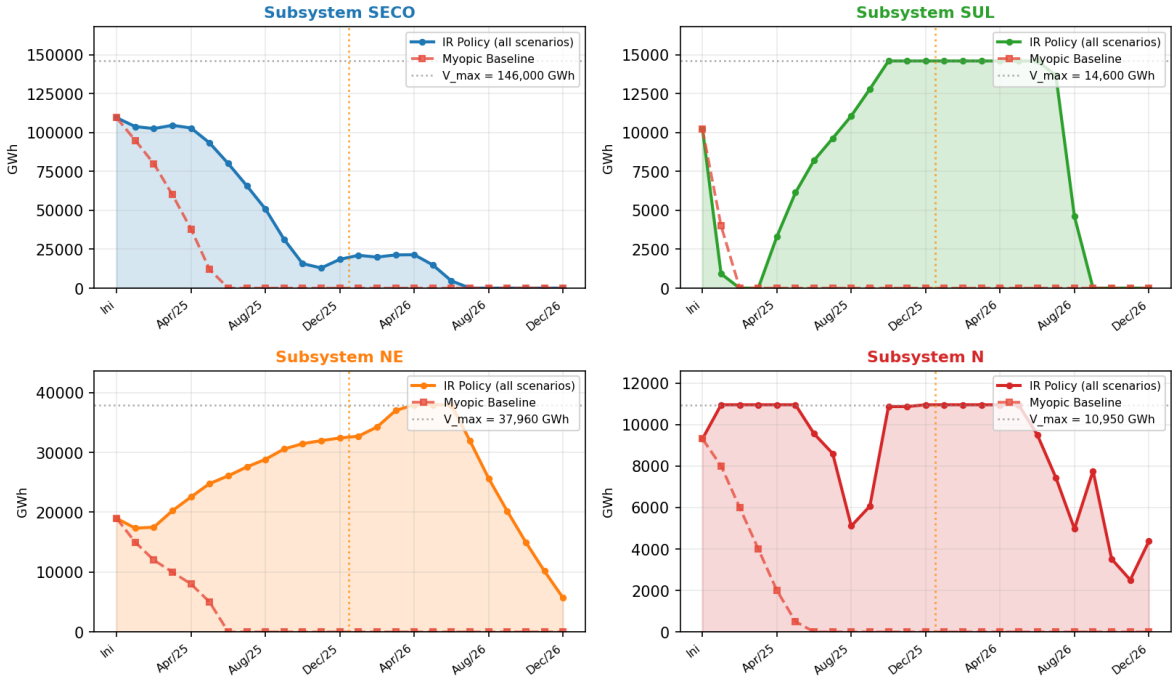
The reasoning loop
plan > act > observe > reflect > repeat



AI backward-induction policy
139.2 M\$

SDDP benchmark
135.4 M\$

Gap: +2.8 % above the benchmark



Reservoirs are preserved during wet seasons instead of being drained too early

Key Takeaways from the Experiment

Caveats & limits

- No pre-programmed dynamic programming solver was exposed to the agent — the simulator worked as a point-evaluation oracle
- The experiment might have limited scalability
- No optimality guarantees: the search is hypothesis-bound — the reasoning depends on the hypotheses the agent explores

The opportunity

- + We can place an MCP at the heart of our model, exposing this structure to the agent
- + The agent can then explore hypotheses through reasoning, orchestrating models and optimization at the core of the workflow

The point is not AI *instead of* algorithms. The point is AI *orchestrating* **algorithms and models, interacting with optimization at the heart of the workflow**



Autonomous Energy Workflows

CLAW: CLI Agent Workflow

CONCEPT

An autonomous orchestration layer that monitors events, delegates tasks to specialized agents, and executes actions through tools and workflows

1 Event Monitoring

Monitor external systems and incoming events such as emails, messages, or API triggers

2 Agent Delegation

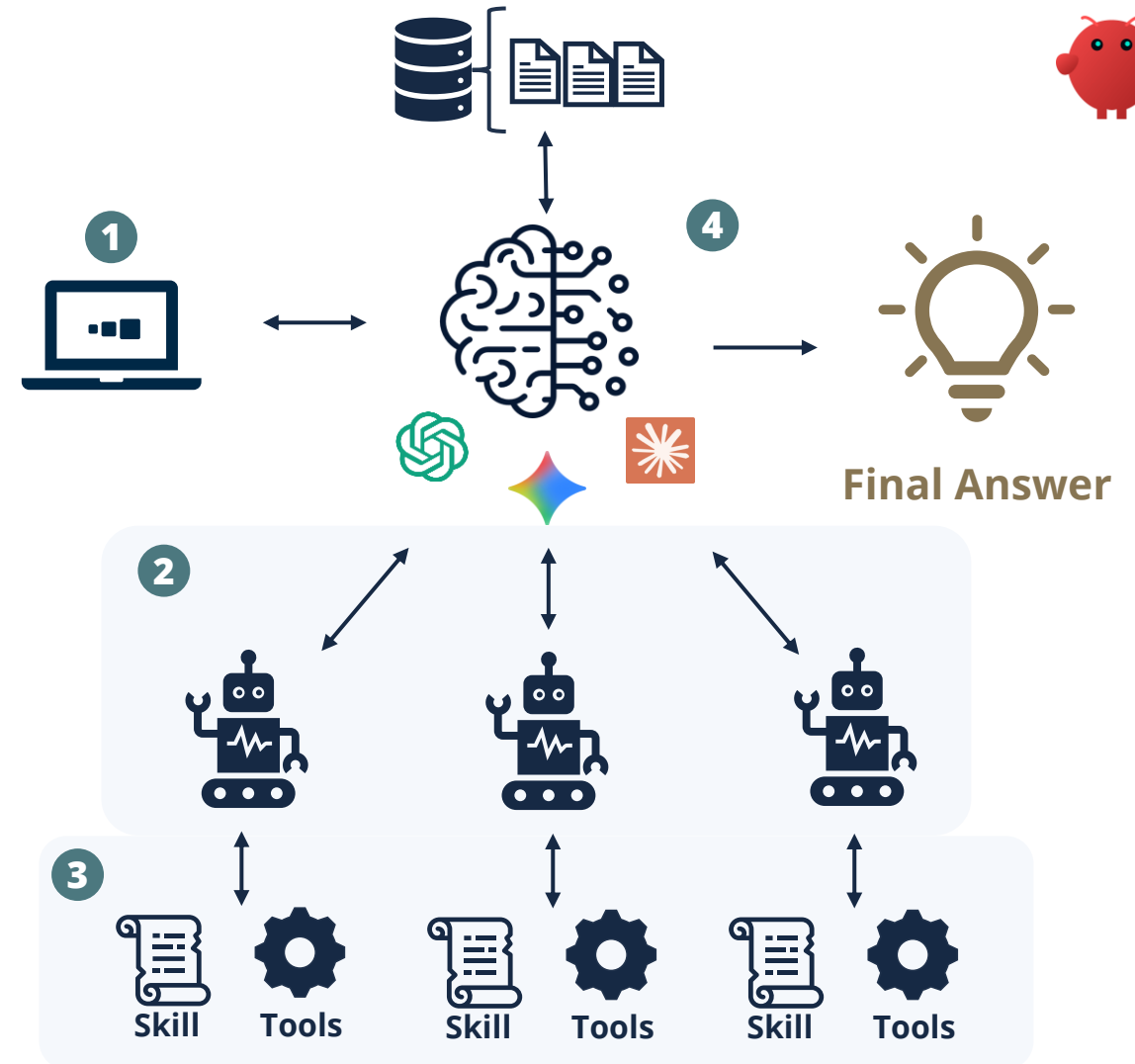
Route the task to the most appropriate specialized sub-agent

3 Skill & Tool Execution

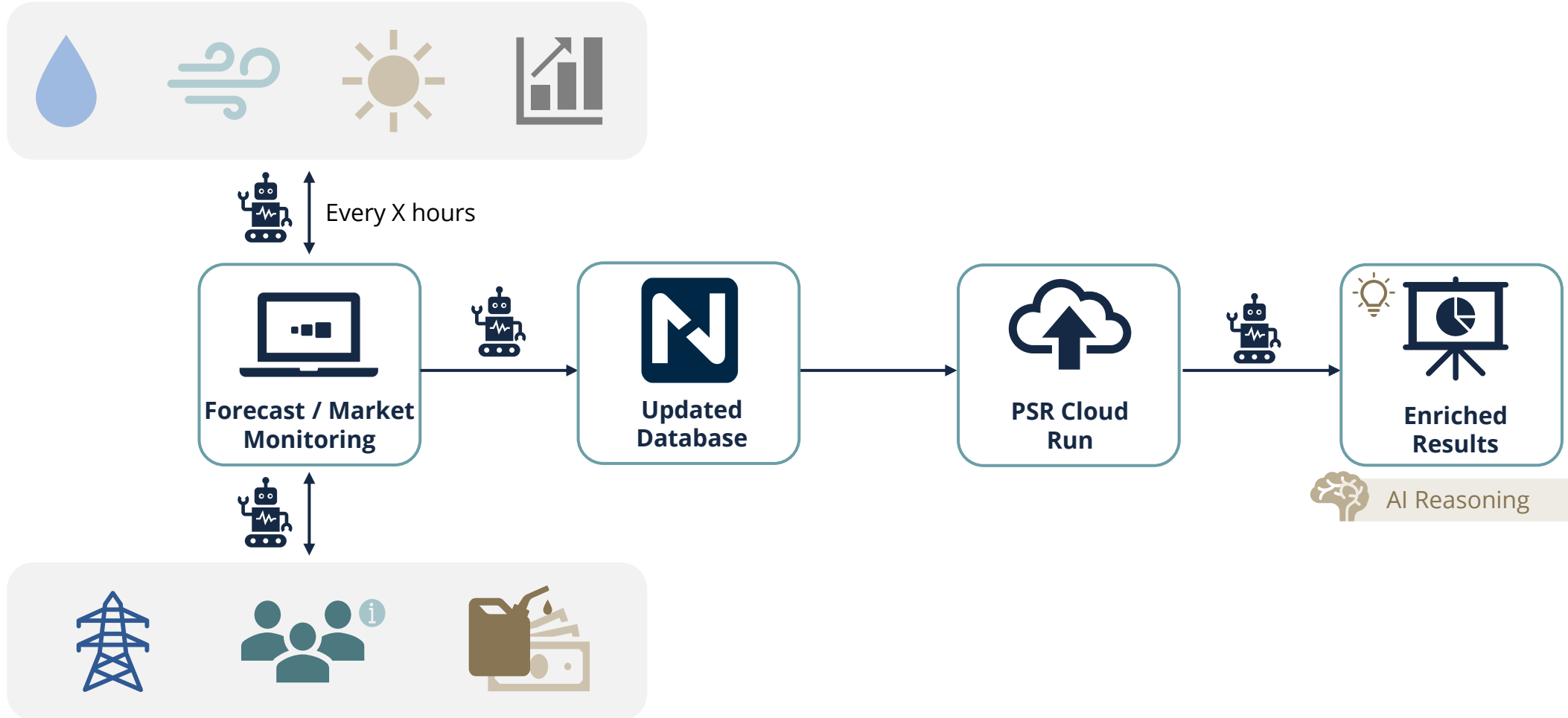
Agents load Skills and invoke external tools through MCP

4 Action & Response

Execute actions and generate the final outcome or response



Autonomous CLAW for the Power Sector



THANK YOU!

Any questions?

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